

Deep Inference with Karl Friston

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Anna Choromanska: Welcome to the seminar series on Modern Artificial Intelligence and the NYU Tandon School of Engineering, where we are bringing together faculty, students, and researchers to discuss the most important work of AI. Anna Choromanska: This talks are being viewed by people around the globe via live stream and we are thrilled to help spread the word about the amazing work going on in the gay community. Anna Choromanska: i'd like to thank Daniela MacArthur batch as well as my own department of electrical and computer engineering for supporting the series. Anna Choromanska: And Russia sunny hostin our esteemed speakers, all of whom are world renowned experts whose research is making an immense impact. Anna Choromanska: On the development of new machine learning techniques and technologies, our guest today his car for someone. Anna Choromanska: Who serves as the principal research fellow and scientific director and the Wellcome trust Center for neural imaging at the University College, London and a professor at the school Institute of neurology. Anna Choromanska: As well as an honorary consultant, and the national hospital neurology and neurosurgery he has been called the genius neuroscientist who might hold the key to true Ai. Anna Choromanska: And authority on brain imaging he earned it a description, with his invention of statistical parametric modeling. Anna Choromanska: A computational technique that allows for more accurate comparisons of activity within different crank out. Anna Choromanska: As well as his development of an emerging technique, known as the box or based more on mattering which you may have read about in a connection with the now famous study of London cab drivers. Anna Choromanska: he's also responsible for dynamic causal modeling a framework that has proven exceptionally useful to neuroscientists. Anna Choromanska: Has a mathematical contributions include the variation of flushing prostitutes and generalized filtering for karate ovation mobile conversion and his main contribution to theoretical neurobiology is a free energy principle for action and perception. Anna Choromanska: Educated at Cambridge University and king's college London, he also holds honorary doctorates from the University of Missouri roundabout university and the University

of New York. Anna Choromanska: You will be here for much longer than we actually have it by Mr all because other owners, but i'll just mention. Anna Choromanska: The first young investigators award and human brain mapping, they may never Gordon brain award. Anna Choromanska: A college de France metal a well known name or your pricing model, the Charles branch Award for unparalleled breakthroughs in brain research and the glass brand award a lifetime achievement award in the field of human brain mapping. Anna Choromanska: Carla is a fellow of the academic medical sciences, the academia European. Anna Choromanska: And both the Royal Society and the Royal Society of biology, as well as a member of emp or an organization dedicated to excellence in the life sciences and I hope that I want to embarrass him by mentioning that he has an H index nearly twice that of Albert Einstein. Anna Choromanska: he's currently working on models of functional integration into human brain and the principles that underlie neuronal interactions i'm excited that he will be speaking to us today. Anna Choromanska: About deep end brands and exploring the fascinating question of whether there is an underlying information theoretic imperative that renders perceptual inference and decision making two sides of the same coin, without further ado let's give him our. Anna Choromanska: call the stage is yours. Karl Friston: Thank you very much for that lovely introduction that was a very complete that was more but it wasn't my CV that's very impressive, I have to say you've mentioned every prize i've ever got that's not just. Karl Friston: A selection. Anna Choromanska: very glad. Karl Friston: I should also say that i've probably i've just spent the past hour, talk, talk. Karl Friston: talking to the students and I I think i've said everything that I know, but now I shall try to reprise some of the key points over over the next few minutes i'm going to be talking. Karl Friston: About deep influence and, by which I mean really. Karl Friston: All the issues that attend self organization sent into behavior and that can be formalized in terms of a normative or. Karl Friston: An optimization like procedure and i'm going to be talking about the imperatives dissenting behavior. Karl Friston: In terms of self evidencing and i'll try and explain what I mean by self evidencing by reference to normative models and. Karl Friston: motivate that particular optimization function that's going to be an evidence lower bound or variation free energy. Karl Friston: From the perspective of those people who might be working in machine learning and artificial intelligence and then i'm going to. Karl Friston: go through some very simple worked examples, just to events, the the basic mechanics of the belief updating and inactive aspects that follow from this treatment. Karl Friston: with a special focus on gathering data, you know gathering data in the service of looking for evidence for my own existence, hence the. Karl

Friston: Rather poetic. Karl Friston: Title from from philosophy of self evidencing so i'm going to conclude with a brief simulation of. Karl Friston: influence into deep gelatin models that have a sort of separation temporal scales and one could think of as a metaphor for the way that we make sense of. Karl Friston: Written or heard language and so self evidencing i'm going to assume at the beginning that we can understand everything in terms of. Karl Friston: The behavior and the internal belief updating of a sentence agent, and then the perception of action as an optimization process and experimentation process and the quantity that i'm assuming. Karl Friston: This maximisation applies to is simply the probability of exchanges with one's world or observable outcomes, given the kind of model that we have in mind or given me, for example. Karl Friston: And i'm doing that i'm expressing it in that way, just to make a connection with many frameworks, with which you will be. Karl Friston: very familiar from one perspective or the other, so, for example, if you're in psychology or engineering. Karl Friston: You can think of the log of the probability of the kinds of outcomes now I would expect to encounter if I was me. Karl Friston: As scoring the attractiveness of the value of those outcomes or states of being that characterize that characterize me and we'll see later this is. Karl Friston: Lower bounded by this variation of free energy a function of observations and some sufficient statistics about the causes of those observations, so if I read this log probability. Karl Friston: As value, then we can sort of shake hands with things like reinforcement learning optimal control theory and engineering and if I was in The Economist expected utility theory. Karl Friston: And an alternative perspective is afforded by information theory, so the negative of this quantity is known in information theory is self information. Karl Friston: Surprise alarm or simply surprise so it's just the implausibility or something like me encountering these outcomes or observations and. Karl Friston: That perspective and lends itself to interpretation of the informatics principle, the principle of maximum mutual information transfer. Karl Friston: closely related notions of maximum efficiency or minimum redundancy by correspondent for example, or indeed an explicit optimization like summarization is free energy in the free energy principle. Karl Friston: that's Nice because the average of surprise is entropy, which means that if i'm trying to maximize in the moment my value or minimize my surprise i'm inherently. Karl Friston: Trying to upper bound or minimize the entropy of my exchanges with the world and that of course is the holy grail of self organization and cybernetics in physics, this will be known as energetic so the Herman Hakan sort. Karl Friston: with a special focus on that template separation temporal scales and if there's a physiologist it's just a statement of homeostasis keeping the variables that matter

within bounds to offset or counter or bound their dissipation and dispersion into non viable regimes, there is a final. Karl Friston: interpretation of this quantity that i'm going to leave, which here, which is that a statistician would take on this quantity, he would. Karl Friston: understand this as a possibility of some observed data and a model that generated those data, and this would then be. Karl Friston: The marginal likelihood or the model evidence, so this just says that I am in the game of maximizing the evidence for my model of the world, and from that we can understand. Karl Friston: The motivation behind things like the blazing busy brain hypothesis evidence accumulation and its various flavors including predictive coding so that's where i'm starting, so let me just drill down on. Karl Friston: This evidence bound this this variation of free energy and see why it is such a useful construct from the perspective of. Karl Friston: artificial intelligence, particularly but also statistics and for those of you. Karl Friston: Doing deep learning with version or to encode, as this is just the elbow or the evidence lower bound and can always be written and as a functional of a. Karl Friston: Effectively, a generative model or beliefs, about the relationship between the consequences observable consequences of some latent states etc particular time, and that is used to optimize my beliefs Q about the hidden or the latent states so here i've expressed the free energy. Karl Friston: That can be thought of as a proxy for the evidence of the marginal likelihood as a negative entropy and so negative energy and the entropy i've done that, just to sort of highlight. Karl Friston: The connection with james's maximum entry principle of you know, the best beliefs about something subjective constraints on those which leave your options open and maximize. Karl Friston: The entropy of those beliefs So what are the constraints they usually articulated in terms of a generative model that comprises a likelihood and the prior the probability of some outcomes, given their causes, and my prior beliefs, about the possibility of those causes and. Karl Friston: This deconstruction of this this way of writing down the objective function behind self evidencing lends itself. Karl Friston: To an interpretation which provides a clear motivation for why you might want to use this kind of objective function and artificial intelligence researcher so, for example, i've written it in a form here that licenses its interpretation that as an evidence bound and. Karl Friston: So i'm expressing the free energy here as an expected log evidence. Karl Friston: plus a kullback divergence, which can never be less than zero and a lower bound and where the log evidence, because it's predicated on a generative model necessarily endows any. Karl Friston: belief, with an explain ability in reference to that generative model so that provides for free so transparent prior assumptions all sorts of technical benefits in terms of hyper parameter optimization of the like what we will

see later, is it also affords. Karl Friston: another kind of optimize it and other kind of days in ophthalmology that is normally associated with the principles of optimal Beijing design. Karl Friston: Basically, getting the right kind of data to resolve uncertainty about how you think there's data were generated and that's in itself important if you're. Karl Friston: Designing machines or artifacts that need to be. Karl Friston: Smart in relation to the kind of data that they operate in and they that they select to induce their belief updating. Karl Friston: Of the problems of the sort, you are getting data forging and empty stomachs and presumably in artificial general intelligence. Karl Friston: i've rewritten this expression here and a slightly different way just by swapping around a couple of terms to express it. Karl Friston: in a form that a statistician will be more comfortable understanding it basically as a mixture of accuracy and complexity, so the accuracies is expected log likelihood. Karl Friston: of my data observations under some beliefs, about how they were caused the latest states. Karl Friston: That is supplemented by a complexity cost, and that is just the divergence the cowl divergence between my prior beliefs of my post your beliefs, having observed the data. Karl Friston: So literally scores the degree to which i've changed my mind after i've seen some data and, of course, the greater degrees of freedom that i'm using up in changing my mind, the greater the complexity cost and this underwrites. Karl Friston: Or is a formal statement of outcomes principle in the sense that we want to supply accurate explanations for observations that are as minimally complex as possible or as are as as simple as possible and what can also read this from the point of view of engineering and that this. Karl Friston: complexity cost is literally going to. Karl Friston: be a computational cost. Karl Friston: For the description of rationality under computational cost constraints in terms of bounded rationality we move from exact bayesian inference to approximate basic inference. Karl Friston: That allows us to express our models in terms of prize and simplifying heuristics that suppress or finesse that computation of cost and if you're a physicist. Karl Friston: Electrical engineer then via the Julian ski quality and Landau is principled that belief updating that destruction of information that is necessary in order to change your mind. Karl Friston: Costs energy, so you can measure the computational costs in terms of the metabolic past or the amount of electricity you're using up in doing you're doing you're pressing. Karl Friston: The real point of this talk, though, is to take that notion of self evident saying and that particular focus on a variation of free energy or an evidence lower bound. Karl Friston: And that's what implications does that have when I am in charge of gathering the data. Karl Friston: Basically, what are the imperatives that emerge when thinking about the consequences of action in terms of the evidence bound or the free

energy that would ensue after I have made an act. Karl Friston: Because we don't at this point, have the outcomes of that action, so we have to take an expectation. Karl Friston: Technically, under the postoperative density over the outcomes, if I did, that and this story, I want to tell her everything else is basically telling the story. Karl Friston: And the story is that when you take this expected free energy they expected accuracy. Karl Friston: Now becomes exactly the objective function that you'll find an optimal Beijing design basically designing the right experiment, the right X the right way or sampling the world. Karl Friston: That maximizes the information gain and it can be regarded as the expected accuracy in a very similar way that computational complexity cost and when. Karl Friston: We take the expectation under some yet to be observed data becomes exactly the objective function found in Beijing decision theory. Karl Friston: The base in risk, essentially, and these two, if you like, parts of the expected objective function are subsumed under unexpected free energy that. Karl Friston: underwrites the optimal choice of behavior or decisions or choices or active sampling and so just to unpack that a motivated in relation to. Karl Friston: Objective functions and normative theory is that you may be more familiar with and I usually start by asking the audience imagine that you're. Karl Friston: out and that you're hungry and then I asked usually the person in the front row, and what are you going to do and I won't embarrass anybody by doing that now, but. Karl Friston: Normally I received the very correct response well i'm going to search for food i'm going to look for food and the answer is absolutely right and here's the vertebrae searching for its food in the answer, there are some really key insights key to. Karl Friston: deciding what kind of objective function, you might look for in terms of trying to understand centered behavior. Karl Friston: So I am deliberately now going to introduce a dialectic between that state value functions of states and. Karl Friston: functionally or beliefs about states and in order to compare and contrast with a qualification i'm going to put them back together again, but for the moment. Karl Friston: let's just think about the standard way of doing this that I could, if I were equipped with a value function. Karl Friston: Of the states of the world that would ensue if I took this action you then I could, in principle, work out the best thing to do, from any given state, and that would be a state action policy. Karl Friston: However. Karl Friston: What we've just said is that the first thing that i'm going to do when faced with a real problem is search. Karl Friston: search for the solution that's searching for the solution entails a resolution of uncertainty about the solution in this instance, the location of the prey. Karl Friston: But uncertainty as an attribute of a belief, and if you allow me to read belief as a probability probability distribution or Beijing belief, then what that tells you is

that the. Karl Friston: The right kind of objective function has to be a functional not have not a function of states, but a function of beliefs about states and i'm noticing that with. Karl Friston: g here so really what we want to do is take the actions to optimize my beliefs about states of the world, furthermore, it matters. Karl Friston: Whether i've searched for the prey and then eat it or whether I try to eat it and then look for it, so we have this added temporality here, and this. Karl Friston: Then we're talking about sequential policy optimization and we have to assess the path of the time integral, of whatever objective function, we choose. Karl Friston: To express it in the form of an action at a time interval of an energy function here. Karl Friston: And the reason that i'm expressing love it like that is that allows me to so compare and contrast two different sorts of principles that you might appeal to when trying to. Karl Friston: formulate articulate and send him behavior so, on the one hand, if you believe that this is how behavior works, then you're interested in subscribing to base balance ophthalmology principled. Karl Friston: And you will find this underneath optimal control theory down deep reinforcement learning expected utility base and decision theory and and so on. Karl Friston: On the other hand, if you think that you're trying to optimize a functional of beliefs, then you're going to subscribe to hamilton's principle stationary action knowledge action. Karl Friston: And examples here include if the energy principle in Europe robotics you'll come across things like intrinsic motivation. Karl Friston: Basically, artificial curiosity, that is there, it is in play, simply to optimize our beliefs about something essentially minimizing our uncertainty. Karl Friston: We know how to make moves sample the right kind of data, so we have baked into this principle of least action. Karl Friston: The principal's office in Beijing design that we can apply to sequential policy optimization dealing with a kind of uncertainty that inherits from partially observed problems such as partially observed mark off decision processes, so what is that G well i'm going to. Karl Friston: sell the G as the expected free energy and and in so doing in juice this computational architecture that can be understood. Karl Friston: us to understand the actual perception cycle, so the idea is very simply, that you have some observable outcomes you. Karl Friston: take these outcomes and you optimize your update your basic beliefs Q about what calls those outcomes, the hidden layton states external states of the world and beyond your sensory veil and and in so doing, you are going to maximize your evidence lower bound or your veil of reality. Karl Friston: And we can regard that as perceptual influence this now affords an optimal belief about hidden states the world that I can then use to roll out into the future and evaluate the expected free energy, and I would get. Karl Friston: If I pursued this particular sequence of actions or policy here, and then I

simply associated probability that i'm pursuing this policy with the. Karl Friston: The expected free energy which plays a role of your gifts potential. Karl Friston: And then select the appropriate action from the selected policy, so this effect, provides the machinery for policy selection I make that act on the world I change the external States they deliver new outcome, and so the circle continues. Karl Friston: So what I want to do now is just unpack the relationship between. Karl Friston: free energy and expected for allergies using these notions of complexity and accuracy in a bit more detail and again what you will see is that these this unpacking discloses things that you will already be quite familiar with so. Karl Friston: What i've done here is is write out the expressions for free energy biracial free energy and expected free energy that we've seen in previous slides just to just to label. Karl Friston: These. Karl Friston: distinct licensed speaker labeling so here's the expression that lens the. Karl Friston: That would be appropriate to understand, this is an evidence about his and log evidence is that the kale divergence that makes it into a bound. Karl Friston: rearranging is moving these two terms around I get my accuracy and complexity and the reason i've done, that is to highlight this beautiful symmetry you get. Karl Friston: The extra free energy, which is just the expectation and the the post area beliefs, about the things that I might encounter observations, if I took this particular policy or decision pie here. Karl Friston: And what happens is that the expected complexity of accuracy become risk and ambiguity and expected kaun divergence and log evidence becoming intrinsic and extrinsic value. Karl Friston: So what do I mean by these things well let's just first of all, ignore extrinsic value and focus on this intrinsic value here. Karl Friston: So what is this well effectively it's a chaos diversion divergence or an expected kale divergence between my beliefs about state of affairs out there, if I were able to see the outcomes consequent on a policy relative to my beliefs, without that so in effect it's the information gay. Karl Friston: afforded by that policy in terms of what I see on my observations consequent upon that policy and in the visual search literature has is known as babies in surprised by people like kitty and bolding more generally, in the context of. Karl Friston: The principles and maximum efficiency it's just the the conditional mutual information between the causes states s and the consequence of observable consequences in the future, given a particular. Karl Friston: policy or sequence of actions, so it scoring technically the relative entropy and the information game, the value of information that ensues from this particular policy. Karl Friston: Sometimes, I repeat, known as intrinsic motivation in the robotics literature. Karl Friston: let's now take various bits of uncertainty off the table and see if we can get back to the the objective function that would operate under balance

optionality principle, so what kinds of uncertainty, can we can we take away well the first kind of uncertainty, we could remove. Karl Friston: Is the uncertainty that inherits from systems that are only partially observed so i'm not going to make the assumption that. Karl Friston: I can actually see everything in the world, so my outcomes my observations my senses can measure exactly everything that matters, so the owners and the SS become same thing, in other words, there is no ambiguity anymore about the. Karl Friston: about the the state of affairs or the consequences of looking at those state of affairs with my observing my my. Karl Friston: My wrist observing operators and then i'm less left with this expression here, which is this account divergence between what I anticipate. Karl Friston: state of affairs or outcomes to be under the policy and my prior beliefs, about the kind of states or outcomes I as this kind of creature. Karl Friston: would anticipate so remember these are the things that are attracted for me that characterize me and often in my world that they'll be referred to. Karl Friston: As prior preferences, so this is just scoring the risk of diverging from my prior preferences under this policy. Karl Friston: and engineering it's known as kate control and in economics it's known as risk sensitive control and it's simply a measure of the goodness of a policy. Karl Friston: and measured in terms of the divergence from your prior preferences and so finally let's just take out another. Karl Friston: source of uncertainty, which is the uncertainty of the cruise because of I don't know the consequences of my actions so basically and now taking the risk part. Karl Friston: Of the the the intrinsic value part of the table just to reveal the extrinsic value here and what is that well it's just the expected utility from the first slide. Karl Friston: That I can read as an expected value, and this, of course, is synonymous with expected utility theory or expected value and reinforcement learning and. Karl Friston: So notice that attempt to get from a sort of principle of least action to a bellman ophthalmology principle what i've had to do is successfully move sources of uncertainty, so that the belief based aspect or the. Karl Friston: The use of a functional beliefs becomes less important than just drilling down on the expected value. Karl Friston: As scored my by prior preferences, so what i'm saying here, is it the bellman optionality principle. Karl Friston: Is a very beautiful thing, but it is a special case of a more general problem when you're operating under different times of uncertainty, expressing that and another way self evidencing in terms of optimizing this is maximizing expected free energy. Karl Friston: In my world known as active influence just is the right mixture of maximizing the expected value and the information game. Karl Friston: That is just a reflection of two sides of the same base basie and automatic coin. Karl Friston: Namely, making the right kinds of decisions but also getting the right kind of data in order to minimize

the uncertainty that enables me to make the right kinds of decisions. Karl Friston: So this cartoon that in terms of expected value pass information game equal expected free energy so i'm going to conclude now and with a couple of simple simulations to just show you how this mechanics on packs in practice. Karl Friston: starting off with a simple observation that in order to quantify the free energy and and particularly the free energy gradients that drive the belief updating and I have to have a generative model. Karl Friston: So this is all about basically inversion active version and regenerative model and and the kind of models which we use to simulate sort of. Karl Friston: trials and behaviors of an experimental so would be a mark of decision or partially observed mark of decision process that would be suitably explained or modeled in terms of this. Karl Friston: graphical model here, where we have states that generate outcomes through a likelihood mapping usually denoted by. Karl Friston: A likelihood matrix or tensor here a and then the dynamics of the world. Karl Friston: are going to be encoded by probability transition matrices b that of course depend upon my behavior the policy that in turn depends upon the expected free energy of that policy G here so that's the structure is coming out as a prize, if you like. Karl Friston: This part of the likelihood of a generative model here the equations and accompany this particular graphical model here we can ignore the moment. Karl Friston: The reason i'm writing this down as a graphical model is to make an interesting point now. Karl Friston: If I can write down the probabilistic graphical model, if I can articulate the mechanics on the point of view of a generative model. Karl Friston: Then the always exists in a company in fact a graph like a conjugate of this kind of model. Karl Friston: In terms of message passing where For those of you are not familiar with photographs basically the. Karl Friston: The the random variables, it says sufficient statistics now mapped to engines and the probability factors now become the modes and the beauty of this. Karl Friston: Is kind of factor graph is that it immediately reveal the kind of message passing, that will be implicit. Karl Friston: If I had this graphical model so, in other words, if I can write down a generative model, and I can always write down the fact graph if I can write down the factor graph I can always. Karl Friston: write down the message passing implicit in modeling version, and these are the messages that have passed amongst various edges here. Karl Friston: giving you a computational architecture that looks very much like what actually happens in the human brain, or indeed any brain and it's remarkably simple, in the sense that you know we only have to infer. Karl Friston: A small number of unknown, these are basically unknown state of affairs in the world what i'm doing my policies. Karl Friston: And, and the model parameters that encode, and this is the initial states D , but more

usually likelihood mappings on the transit transition matrices A and B respectively and. Karl Friston: And if one just takes off the shelf variations message passing derived from that factor graph. Karl Friston: organized in a quarter that paragraph, you get update equations very much like the kind of equations that people use to simulate. Karl Friston: Your own dynamics and plasticity in the brain so, for example, the expected states of the world, under a given policy is just a nonlinear say voltage current activation function of some linear mixtures of. Karl Friston: represents a neuronal representations of outcomes and beliefs, about the past and the future. Karl Friston: Policy selection just becomes a standard soft Max function of my expected free energy here I just equipped. Karl Friston: This free energy when the temperature of precision parameter that itself can be optimized with respect to. Karl Friston: free energy and learning corresponds to updating the model parameters here the initial status just been counting updating. Karl Friston: Jewish life accounts look very much looks very much like assisted Pasty, and having classes in the brain and action selection is just once applied my soft Max function I just choose the action that has the. Karl Friston: greatest expected value or expected free energy, and one can play games in terms of working out which parts of the brain this might happen. Karl Friston: But what I want to do is just briefly show you a demonstration of this in action ease and the simplest model possible God model possible and so in this generative model we're simulating. Karl Friston: A little to step maze tasks, so the idea is that this. Karl Friston: Little mouse here can make two moves and and its objective is to secure its prior preferences arts rewards which could be on the left or the right hand side. Karl Friston: The twist here is that there's this instructional queue but if he uses up its first move to go to see the color of this instructional Q this Q will tell it where the reward is so it can guarantee getting the reward on the second move. Karl Friston: And I should say that once it's made a move, it has to stay to one of these two arms here, these are absorbing states, and it has to stay there, which means that. Karl Friston: From the point of view of expected utility. Karl Friston: That has the same expected utility as taking a gamble and seeing if it's on the right or the left and then having to stay there. Karl Friston: half the number of times, it will enjoy being with its food on two moves or to eat parks and the other half the time there will be no. Karl Friston: No reward at all so from the point of view of the webinar technology principle both policies all combinations of passes on those that involve. Karl Friston: Visiting the instructional queue and then the reward or going to a putative reward location and staying there they both have the same expected utility but from the point of view of self evidencing clearly it's going to be more useful to resolve your uncertainty. Karl Friston: By going to the

instruction queue and then go and getting your reward and that's exactly what happened so i'm just illustrating the characteristic behaviors you get this kind of behavior yo whatever example you choose, but you is what one of the simpler and. Karl Friston: original examples we use to demonstrate this, so what i'm showing here is the behavior in terms of the. Karl Friston: Initial state whether the reward was on the right or the left hand side of a sequence of trials, the. Karl Friston: action or the policy that was selected one of 10 combinations of different moving up and down was going to the right or the left and, as time goes on the actual performance. Karl Friston: In terms of the expected utility or negative cost and the outcomes and, crucially, we equip this agent with the ability to learn. Karl Friston: The initial context from at each trial was the reward on the right or the left and initially we switched it around, but then we left the reward on the left hand side. Karl Friston: So that over trials this agent learn that the reward is always on the side so initially it goes and does is epistemic porridge in response to this epistemic avoidance of this intrinsic value this information gain the value of information. Karl Friston: And then confidently goes to secure its reward but as time goes on. Karl Friston: That epistemic affording so that intrinsic value starts to wane why well because it learns that the food is always over there and there's little information. Karl Friston: value in wasting a move by going to the instructional queue or you can read this as a condition stimulus so at a certain point there's a threshold crossing where the extrinsic value. Karl Friston: becomes greater than the intrinsic value that is slowly reducing as the as the agent becomes familiar with its environment and less and less curious about. Karl Friston: The state of affairs and respond to less and less to the epistemic avoidance of this of this condition stimulus at a certain point it switches to. Karl Friston: This exploitative behavior So what we have here is a really simple model of a deterministic transition from explorer exploration to exploitation. Karl Friston: That is a simple reflection of the fact that there is belief updating in the game and a. Karl Friston: Certain point that Beijing optimal design or that epistemic part of the objective function is now given way to the pragmatic part. Karl Friston: That is scored by the expected utility or the expected realization of the prior preferences and so i'm going to conclude now with. Karl Friston: Exactly the same story, but in a slightly more elaborate setting where i'm going to completely ignore reward and extrinsic value and just focus on this epidemic avoidance. Karl Friston: Part of the expected free energy this uncertainty resolving aspect and seeing how far we can get in simulating something which were very expert APP but it's actually a really challenging thing, which is, as you collect humiliating evidence from spoken or heard or read language. Karl Friston: And abstracting meaning from that at different

temporal scales so let's let's return to this formulation of the task, the problem and solution in terms of. Karl Friston: Active influence as an inversion of a generative model and in terms of from the point of view of the graphical model and the accompanying factor graph. Karl Friston: And now let's just ask well, how can we make this more realistic in terms of going beyond the mark hoeven kind of causality and structure in a in a real and lived world to the kind of semi marker marker of instructors we see with dp structure causality and it's really very simple to. Karl Friston: To put this. Karl Friston: This deep structure into these gentlemen models simply by putting one mark of decision process on top of another one with a key twist that the. Karl Friston: transitions at the higher level unfold more slowly than the transitions at the lower level, so you can think of the higher level as basically supplying or modeling or generating a context in which the lower level unfolds or packs. Karl Friston: And literally in this instance, this journey model generates occasionally a new initial state, which was the context and the previous example of was the reward on the Right or the Left. Karl Friston: And, given that, then we can have states of the lower level fluctuating and transiting and transiting through orbits or not. Karl Friston: And to generate a sequence of outcomes at the lower level where one or other or both are informed by prior preferences via this inspected reality. Karl Friston: So what we have here is a deep in a diet chronic or God model where we got slow stuff going on at the top and fastest stuff going on at the bottom, which generates these first century samples that are all contextualize by. Karl Friston: This slower dynamic the higher levels of the model on the point of view of message passing and that generation top down generation of the context i'm packs into. Karl Friston: or translates in terms of modeling version as a sippy using the evidence from trajectories at a lower level. Karl Friston: to inform the beliefs, about the context in which i'm operating so slowly accumulating evidence in the sense that i'm updating my beliefs that God is slow, changing context. Karl Friston: Based upon the processing of fast evidence accumulation mechanics afforded by this message passing out the lower level of the model and here they accompanying. Karl Friston: Formal definition, so the junction model and the variation message passing it comes from the spectrograph so let me close them with. Karl Friston: Integrating those brief update equations and a deep journey model that has this diet chronic or D temporal aspect and what we've done here is trying to model, a very elemental kind of reading where instead of letters we're using little icons to stand in for. Karl Friston: Words so the at the lowest level the kinds of outcomes that this agent. Karl Friston: Can sample are basically letters of icons here policy it's a bird and a cat it also knows where it is looking at the current word. Karl Friston: And, and to generate

at any one point in time, the outcome or down to see the two sensory inputs, if you like, i'm going to need to know the latest states which is basically. Karl Friston: Where Am I looking in terms of this particular word and what is the word, so if I knew what the word was this composition of letters here. Karl Friston: which will call flee when there's a bird next to the cat, it has to fly away or it can eat or just wait it has nothing next to it, and if I knew the word. Karl Friston: And I knew where I was looking and also i'm introducing extra flip hidden state here which you can think of no change in finance, for example. Karl Friston: And then I can generate exactly what I would see at any one time, and I can just write that down to in a very simple mock of decision process, depending upon where is a card to. Karl Friston: That changes the controllable stage, which is where on this in this word i'm currently directing my gaze but now I can equip this model with an extra level. Karl Friston: And so what would I need to learn to know what word I am currently looking at well I need to know the context supplied by the sentence from which this word came. Karl Friston: So here in this example or six sentences that unfold, as I move from one word to the next, in the sentence and in a particular order, depending upon which sentence i'm currently looking at. Karl Friston: So, again I need to know where where i'm looking in terms of which word i'm looking at and what sentence i'm looking at it, if I know where i'm looking at what i'm looking at. Karl Friston: Then I can tell exactly which word and currently scrutinizing or sampling of origin visually painting. Karl Friston: And then I can generate given the lower level states, the actual outcomes and I can also generate outcomes in terms of. Karl Friston: Where we are which page i'm looking i'm looking at so again a very, very simple generative model we now take a factor graph we operationalize as brief updates integrating the system supplying. Karl Friston: The appropriate stimuli to this agent, who now is actively in charge of gathering gathering the next bit of information in terms of where she looks. Karl Friston: And what on this particular letter and which letter to jump to in the future, so this is sort of epistemic foraging of a pure sword. Karl Friston: That is meant to simulate the way that we scan text when reading, and indeed the kind of behavior that. Karl Friston: emerges from this formulation is very reminiscent, to the well we actually read things in the sense that well, let me just show you what describe what this is showing. Karl Friston: What we're seeing here are a sequence the sentence for the sequence of words that say three way to feed wait, which is the what the Center the sentence that this. Karl Friston: Synthetic agent was was reading and here are psychotic i'm movement so initially. Karl Friston: She jumps to the cat and then immediately ignores the rest of the letters in this word and jumps to the next word. Karl Friston: Simply

because she knows exactly that this must be a free it can't be anything else so she doesn't waste any time foraging around the rest of the letter she goes on to the next information rich epistemic really. Karl Friston: valuable and or uses the systemic invaluable action which is actually jumping to the next word and just has a couple of samples here, concluding that this is a way to. Karl Friston: word and then jumps on to the next word and forges around until she gets the the other most informative and definitive evidence that this is a word and then move on to the next one. Karl Friston: So this is a depiction of exactly that simulation in terms of evidence accumulation and belief updating and that's trying to illustrate the implicit separation of temporal scales under these deep generative models so what we've done here. Karl Friston: is just plot the neuro the synthetic neural activity so encoding expectations about what is the word the letter, the time or the word that i'm currently. Karl Friston: Looking at given my sampling of the letters and and what is a sentence that is generating that letter. Karl Friston: Now these so that word, so the first level, there are only three words that I could possibly bring to the table to account for my sensory input. Karl Friston: And these change, depending upon which letter i'm looking at, and we see rapid changes in the blue to flee weight feed and ultimately weight and and, of course, that is, the sequence that has been generated by the top sentence here for the free way to feed and wait. Karl Friston: And, as I am. Karl Friston: Building these beliefs optimizing these beliefs and choosing the actions that. Karl Friston: optimize those beliefs or resolve and certainty about those beliefs and i'm also accumulating evidence about which sentence, I am actually reading. Karl Friston: Now notice the first and the four sentences are distinguished by the last words, I can only resolve my and bivalent about whether i'm looking at these two sentences right at the end when I see the last word, which is indeed wait. Karl Friston: This structure of shows you exactly the same data from both levels but we're simply filtered it using band pass filtering do you typically use electrophysiology to generate synthetic field potentials level field potentials that allows. Karl Friston: That allows us to make an observation that the the basic mechanics in play here. Karl Friston: generate belief updating schemes that look very much like what you see in a real brain so, for example in terms of unit activity, you can. Karl Friston: Think of these representations here expectations as being reflected in Paris carry delay productivity in the prefrontal cortex and there's a remarkable similarity between. Karl Friston: The synthetic er PS or local field potentials and, paradoxically, feel potentials and Jerry active vision in in sort of lower and higher level. Karl Friston: Levels of the visual of the visual hierarchy here speaking to a certain your own plausibility needs this coarse grained level of analysis. Karl Friston: Of the

kind of message passing, that is implied by these by this this generative model so that's the story and i'm going to give the last word to Einstein in terms of making things simple just to emphasize that all that epidemics, all that. Karl Friston: exploitive. Karl Friston: Information seeking behavior is basically acting in a way to maximize expected simplicity resolve uncertainty that becomes part of this. Karl Friston: This imperative to maximize expected model evidence or it's bound in terms of elbow or variation free energy, so it just remains for me to thank those people whose ideas i've been talking about and, of course, thank you very much for your attention, thank you. Anna Choromanska: Thank you call for very exciting presentation on, we have now some time to take questions from the audience, I know there are some questions on the question and answers. Raquel C Thompson: Yes, let's start with the Q amp a first, and then we have some in the chat as well, but we'll start I think most the have some may have an answer it but we'll start with the Q amp a first so Australia. Raquel C Thompson: We can. Raquel C Thompson: Yes, read it out, and then we can go from there alright. shreya ahuja: there's a question from pepper shaker it says how to set free and the principal ftp lead to a cybernetic analysis of systems, in particular those bit symbolic language use. shreya ahuja: For instance, baidu biological systems with language or culture consumes so much more energy than those without, for example, the famous anthropologist se invite noted. shreya ahuja: Culture was as though almost as the amount of the energy harness per capita per yard is increased or as the efficiency of the instrumental means of putting the energy to work is increased. Karl Friston: Also, an excellent. Karl Friston: If wide ranging question which allows. Karl Friston: which allows me to sort of just pick up on some some key points that are implicit in this question yep first of all question, how does it relate to a cybernetic analysis extremely closely. Karl Friston: So this notion of a generative model that's trying to secure evidence for itself and is very, very closely, in fact, one could read it as a sort of 21st century version of the good regulated theorem. Karl Friston: From Ross Ashby and colleagues, so one of the fathers of cybernetics where the theorem which he considered provable is that. Karl Friston: If you've got any regulator any subsystem in a larger system in which is immersed in this environment any system that can control a regulated environment successfully must be a model of that environment and that's exactly the same assumption. Karl Friston: In this formulation of active influence in that I am a generative model of my little world and that all I have to do is to optimize my model to make it fit for purpose to explain and account for and account for my. Karl Friston: My model and the second point, so I should say, of course, that that's a notion of the good regulator theorem and cybernetics is not so

distant from. Karl Friston: So notions of self assembling computational chemistry, or indeed or two places in the life sciences, which again I would. Karl Friston: I would. Karl Friston: I would say that the formulas and i've been talking about can be regarded as a mathematical statement of of these notions and then the next set of questions are all about. Karl Friston: The consumption of energy and symbolic languages, and I think those speak to some really key points that are unified by this drive to simplicity, the the. Karl Friston: The minimization of complexity and I remember talking earlier this afternoon about your the importance of eluding sharp minima and trying to find this global. Karl Friston: flat minimum that have this maximum entropy this maximum curvature aspect is that you know this closely related to outcomes principle, so the minimization of complexity. Karl Friston: And what's the best way of making your explanation for the world your generative model as simple as possible, you cause brain. Karl Friston: So how are you going to cause brain a very high dimensional continuous state space out there layton states, generating. Karl Friston: Your sensory impressions well, one of the simplest ways is just to Course Credit into discrete space based model so from my perspective, you know, having the language and symbols is just a. Karl Friston: way of saying that perhaps it's more appropriate to have to have discrete space based models, you know sort of silent machines hidden Markov models mockup decision. Karl Friston: Because, whatever you want generating sequences of data, because they have the right level of complexity and then the question comes on how coarse grained do you want your states to be how many states are there, how many words are there. Karl Friston: What kind of like algebra do you want, you know, do you want to reduce your description to. Karl Friston: And so I would look at the emergence of symbolic language use ignoring all the in culturally originally psychology psychology arguments that you can also frame in terms of. Karl Friston: As a model section natural selection when ignoring that and just focusing on the simplicity. Karl Friston: That you as afforded by course reading and categorizing having symbols that standing before in the semiotic sense. Karl Friston: You know what would otherwise be a rather complex game to try and model accurately and so, if it's simpler to use a language or set of symbols of course raining. Karl Friston: Why is it costs so much energy when i've just said that complexity and stores, the the energy costs by then Dallas principle and I think the answer to that is remember the evidence is the difference between accuracy and complexity, so if you're driven to if you're compelled. Karl Friston: To maximize your evidence you're compelled to maximize the accuracy and but keep the complexity. Karl Friston: as low as possible, but the complexity is always chasing the accuracy, so there will be a natural tendency

any system that is adaptive and accumulates information and. Karl Friston: Whose generative model becomes more and more apt to explain it says to exchanges, there will be unnecessary complexity cost that is more than licensed by the increase in the accuracy accuracy until a certain point where you don't go. Karl Friston: Any higher, otherwise you start over threatened and fell to generalize, and so I think that you see the. Karl Friston: A careful and discriminate use of energy in evolution evolutionary context is very reflection of the growing complexity of. Karl Friston: magenta and morals in possibly ourself assembled or niche constructed and in cultured worlds, and that is fully licensed by increasing and accuracy. Karl Friston: Everything is now more predictable and we are better at predicting it, and that is especially true of language and there is just a reflection of this tendency to maximize module likelihood or model evidence. Karl Friston: that's my answer. shreya ahuja: Professor there's another question Should I be it oh. Karl Friston: Yes, of course, so I can read it as well, so i'm cheating to us. shreya ahuja: Okay um So the question is in the factor in the vector graph should there be a cycle between G, the goal of the policy to minimize surprise and the the model being trained or. shreya ahuja: been counting that I think is used to be, then as as for the the agent generated on data to minimize surprises and then train the model to model, the ball and its belief properly. Karl Friston: Right, I have read that one that sells a very sophisticated question so if I do remember properly the question here is it in the sort of. Karl Friston: message passing factor graph formulation. Karl Friston: And should the G be connected to the the the outside and yeah that's that's absolutely right so. Karl Friston: If you remember when we have this graphic talking about the actual perception cycle and I was emphasizing the circular causality that is mediated by the external states that is certainly a really important aspect to this kind of updating work. Karl Friston: Your beliefs about what to do next determine the actual action upon real states of the world that generate the data so that now you've got this curious again almost automatic aspect to influence. Karl Friston: Hence acted in France and where you're in charge your influences determine the data that you're making inferences about and, of course, you know you're you're leveraging that sort of confirmation bias and circularity in keeping yourself. Karl Friston: sampling the kind of data that you want to you want to the sample so there's an inherent, if you like confirmation bias and optimism bias that written into this this kind of circular mechanics that just is the description of self organization to to a particular set of outcomes. Karl Friston: and implicitly as a particular set of states that generate those outcomes, so I hope that was a spiritual the question back I can't actually read it so. Karl Friston: I don't know if the person who was asking the questions is present, or whether you

have to ask them if that was a satisfactory. Karl Friston: Is there something that breaks us out of this cycle. shreya ahuja: question. Karl Friston: i'll answer that question, I can see that popping up on my street so i'm not sure that's allowed to just answer ad hoc questions. Anna Choromanska: Yes, go ahead, I think there's a to follow up from glam Is there something that breaks us out of the cycle, and so we know we can get a perfect model. Karl Friston: yeah I mean there's no such thing as as the perfect model in the sense that you know. Karl Friston: What I haven't spoken about today, the different levels of. Karl Friston: optimizing that next realizing this this free energy functional and so it can be we we don't briefly learning when we're looking at an example of learning the. Karl Friston: Synthetic rattling The context in an environment with which she was becoming familiar and but all the more important aspects or an equally important aspect is is the structure of. Karl Friston: The generative model itself in terms of you know which connections contingencies, how many levels and a hierarchical model, how many factors, do you introduce. Karl Friston: And this kind of basic model selection, which is just another free energy optimizing process where or maximum margin likely optimizing process and is know what could be blank josh 10 and bonus structure learning. Karl Friston: And I recently learned that comes from the field of radical constructivism and i've been exactly the same problem exists or not problem but exactly the same optimization process. Karl Friston: emerges from this treatment of centering behavior so that your model itself is always been optimized either at an evolutionary time scale, so that in. Karl Friston: On that, reading natural selection becomes nature's way of doing basic model selection by extra amazing and fitness function which, in this instance, would be. Karl Friston: A variation free energy and the point being that there is never a perfect model and that, and you know you ask yourself a model of what well of the data that's being generated. Karl Friston: and well, but if you're in charge of your of your data yeah, how do you ever know you know. Karl Friston: What the perfect model is, and of course the perfect model is just you know, in a sense, yeah and when the environment supplies exactly what you predicted and you minimize your complexity and. Karl Friston: Everything unfolds according to your prior beliefs, but, of course, in reality, you're dealing with a capricious changes in changing environment, simply because our environment is built of or comprises not so creatures, like us, so we're always chasing, if you like. Karl Friston: You know the the structure and the parameter ization and the states under a best model where best is this operation defined in terms of of the marginal likelihood given some data so that's. Karl Friston: that's how so. Karl Friston: you'll get to a perfect model, given that perfect model doesn't exist, simply by. Karl

Friston: Simply by learning and doing a structure learning or learning in terms of model parameters say experienced Bennett plasticity neurodevelopmental learning and. Karl Friston: and basic model selection again at a neuro developmental level or an evolutionary level in in the service of maximizing maximizing model evidence. shreya ahuja: Professor there's a related question it says, can you discuss how cognitive conflicts relates to breaking out of the cycle and when is it optimal to engage in it twice is not. Karl Friston: Yes, excellent question it sort of brings to mind some notions of cognitive dissonance and and the like, and so. Karl Friston: I think it's informative you're not gonna answer that question, but certainly it's a great question and you start to ask well how would you simulate that in silica How would you. Karl Friston: How would you actually sort of simulate a base optimal agent who's in a state of crisis or existential angst or cognitive dissonance. Karl Friston: And the way that that normally works is that you and your contended you set up a situation where all the sensory evidence at hand. Karl Friston: speaks to a very low confidence way forward and so we're now talking about the beliefs about different policies or choices are plans you know there's more simply plans. Karl Friston: If it's the case that the two or more plans are equally likely, then the precision of your beliefs over pie or your plans now have a very high entropy and that's usually associated with a kind of dissonance a kind of answered and your and indeed. Karl Friston: A negative Valence that now, you can can be, if you like, registering a state of conflict in this in the limited sense that there now conflicting ways forward that are equally attractive in the sense that you are assigning an equal posterior probability wait to and, interestingly. Karl Friston: The the if you work out how you would optimize the hyper parameter controlling the precision or the entropy of those beliefs, and do so in terms of a. Karl Friston: So it gives distribution, it looks as if domain has all the characteristics that is in the game of registering. Karl Friston: When you're in this state of dissonance or uncertainty about what to do next, and when you resolve that and when you know exactly what you're going to do next, and commit to a particular plan of action very confidently. Karl Friston: And then the precision of those beliefs increases and the hyper parameter encoding that precision increases and and that, and if you simulate it looks very much like sort of your. Karl Friston: Face I mean definitely response so if the question was about your, how do you understand dissonance and conflict and in sort of active infants world it is really about it's put simply it's really the. Karl Friston: The uncertainty that requires you to select among a number of equally plausible hypotheses or models or explanations. Karl Friston: And, in particular when those hypotheses are beliefs, about how you're going to act, and that can have a very personally gentle. Karl Friston: aspect

to it that you know people normally associate with people with anxiety and angst so is that a reasonable answer is that what the question was was getting up to do you think i'm not sure.

shreya ahuja: Professor there's another question and it's from Tony award winner he's josh benji oh and i'm going to read his question now how to make the difference between epidemic uncertainty and the doric uncertainty which are both included in the surprise information gathering policies.

Karl Friston: Right, can you just repeat the two technical phrases have them.

shreya ahuja: The epidemic and sadhana D and the alert orrick and certainly.

Karl Friston: I have to ask him what i've heard that before but you're gonna have to ask him if you can put it in a slightly simpler phraseology for me.

Anna Choromanska: Right, maybe we can unmute your slot so that he can ask his question.

shreya ahuja: yeah I can do that i'll do that, right now, one second.

Yoshua Bengio: Okay, so ultraconservative is just noise it's a the uncertainty in the world that no matter how much you acquire information, you will not be able to resolve, so if you.

Yoshua Bengio: are trying to just maximize the amount of surprise in your quest for information, you might go to places have a lot of noise.

Yoshua Bengio: is really only care about the.

Yoshua Bengio: epistemic uncertainty, which is the reducible part of the unpredictability.

Karl Friston: Perfect Thank you and.

Karl Friston: So.

Karl Friston: yeah I should have done, I have been told that, before I just forgot it.

Karl Friston: And yeah so that impacts in this decomposition of this expected the national boundary expected free energy and in terms of the pure epistemic.

Karl Friston: part, that is, the reduce blood certainty so that scored by the anticipated or the expected information gain scored by the kale divergence greener pasture and the prior.

Karl Friston: And the other editor can't pronounce it, but the noisy gate and the noisy bit is there, but requires you to rearrange the terms, and that would be be basically the ambiguity which may be irresolvable.

Karl Friston: So the noisy bed is usually written in terms and these discrete states based models into the likelihood function so it's the precision of the.

Karl Friston: likelihood of mapping, so that, in a very dark room minor very noisy context or outcomes become equally likely for any given cause so.

Karl Friston: They in the decomposition that I was walking through in the presentation yeah I would assign the noisy bit to the ambiguity which you're trying to minimize.

Karl Friston: And the academic uncertainty, the to the intrinsic value, which is the reducible part which is the reducible part of it, so when you put it all together, it looks as if the agents are selecting those policies.

Karl Friston: that are going to be optimal in in reducing the reducible answer to the systemic uncertainty.

Karl Friston: while avoiding states of being that have very noisy likelihood mappings so that you will, in the absence of any stomach.

Karl Friston: Any any epistemic

value, and you will be basically searching under the lamppost so that you're avoiding noisy or darker and informative ambiguous. Karl Friston: scenarios, or you want simple manifestation of that isn't. Karl Friston: It's an example and that usually comes out in the context of the what's called the dark room problem in philosophy, if you want to minimize surprise, why do you just go to a dark room, so the lights off and go. Karl Friston: and stay there, quietly then you're not going to be nothing you know to expect anything to happen and therefore your surprises minimized, and so it is exactly this. Karl Friston: imperative to minimize expected surprise, by which I mean minimize uncertainty which, if you like, dissolves the dark room problem, because the first thing you do like a dedication does when. Karl Friston: She goes into a room is to switch the light on to make sure that she doesn't bump into things and so that the there's two aspects of different reduce reduce uncertainty is absolutely crucial in shaping the right kind of behavior and under this formalism it's not the kind of. Karl Friston: dead say Allah Allah erratic noise that you're talking about. Yoshua Bengio: yeah so there's an issue with. Yoshua Bengio: uncertainty and noise. Yoshua Bengio: If I understand well you're saying it's going to be captured by some parameters of my model like like you know gaussian variants or something like this. Yoshua Bengio: And that's going to allow us to factor, the epistemic reducible part of the certainty, but the problem is when an agent is. Yoshua Bengio: Building its model of the world based on finite amount of data so in the limit what you say, works, but in general, if you want there's going to be overfitting right so we're going to. Yoshua Bengio: underestimate the. Yoshua Bengio: How. Yoshua Bengio: how badly we will predict in the future because of that, and so it makes it difficult to disentangle epistemic I mean epistemic and record noise. Karl Friston: yep great question we've sort of speaks to two key points so you're absolutely right that in all of these instances realizations demos of this kind of active influence there is an implicit. Karl Friston: estimation of this kind of noise so in the when you're dealing with continuous states based models. Karl Friston: they'd be normally cast on these infants part would be normally casters are basic and filter an extended basically filter. Karl Friston: And the noise estimation part would be the Kalman game and so inevitably in the skills, you have to optimize the hyper parameters that underwriter can gain. Karl Friston: When you look at what that would manifest as in a sort of neural network setting it looks very much like attention. Karl Friston: And so that you know by optimizing the precision of likelihood mappings under continuous state space models during a base in filter. Karl Friston: That looks very much like what the brain does in terms of 40 more attentional gain or weight to certain sensory inputs that it believes have. Karl Friston: A greater precision

and thereby down there with the ability to induce believe updating hiring say. Karl Friston: Across the hierarchy indiscreet states based models, it is literally, as you say, the precision of the likelihood mapping. Karl Friston: Usually written down as Jewish Lee process and that can be now conditioned upon different contexts, so the movie on the second part you're probably you're saying yes, it is very difficult to do that, it is very difficult to do that. Karl Friston: But it has to be done, it is done, and you know my own existence proof and all that and the weather, that is normally done is by now having a model space in which you conditioned the precision of your likelihood. Karl Friston: model on a context where the context is inferred that you're at an appropriate level in in a hierarchical model so you're right in the. Karl Friston: The attentional sampling, as it were, into the journey model and then you optimize that just by belief updating in the normal way or, and this is, you will be learning in the normal way learning the Jewish parameters that described this in respect to you. Karl Friston: to your your natural question yeah but surely this is an impossible problem at the beginning, because of the problem of overfitting and now I really. Karl Friston: Think it's important to sort of push back on that the whole point of using a variation of an elbow function, it is equipped with a complexity penalty. Karl Friston: which precludes over fitting in principle I can't overfed well that, and so I don't think it's the overfitting, which is the problem, I think the real problem which you may have been speaking to is actually. Karl Friston: testing the universe of models that you could bring to the table to properly accommodate the conditioning or say. Karl Friston: The precision of various mappings, for example on different kinds of context and and that you know, I think that that is an unsolved problem, but clearly. Karl Friston: And again, because you and I are here, something some, but some artifacts of solve that problem and and I put that under the under in the. Karl Friston: problem to enter structured learning yeah finding out the right Jeremy and models that can recognize the right context they're in. Karl Friston: And then deploy the right decisions to optimally sample the right kind of data or ignore other kinds of data, and I use ignore here in the psychological sense the compliment of you know, applying. Karl Friston: attentional selection to one subspace of data space or another. Karl Friston: Is that what you meant by the difficulty of doing this before you know the nature of the journey model. Yoshua Bengio: So I mean I would the orbiting part is. Yoshua Bengio: Due to the you know the in the finite data set learners tend to you're saying there's no there's no fitting if you're basing I just don't believe it. Yoshua Bengio: And, and the reason is you don't have a perfect prior you may have like wrong assumptions so there's like bias and that's not going to be accounted for in your Beijing variance. Yoshua Bengio: is going

to be a gap. Yoshua Bengio: between what you think is how well you're doing and you're going to enter so you're going to overestimate how good you are in your predictions. Karl Friston: yep okay well that's two really important points, first of all, and because we deal with hierarchical gentle models, there is no one prior so you know all the. Karl Friston: All the heavy lifting is done by the intermediate levels in a hierarchical gentle model so that empirical prize, so they are updated and they are learning. Karl Friston: So they're learning about trainable prize so there's no, there is no wrong and prior in these hierarchical models, but if there was, and I think there is, if you allow me to. Karl Friston: assign a structure of the model, the Labor prior So if I bring a particular structure, so I brought in your va or. Karl Friston: Again, to the table and it had say six levels and they did or did not have some recurrent connectivity, no, no, all the architectural aspect, if you allow me to call that a prior. Karl Friston: That, I think I know I can see your point is that if you start off with the wrong structure, then you're not going to be you're not going to be able to optimize the module likelihood or the. Karl Friston: Or the the model evidence, but. Karl Friston: So what what does that mean well, it means you have to put structure learning into the game, you have to explore different structures and that's what I was talking before about. Karl Friston: Basic model selection in terms of the structure selection, which brings us back to the problem now of exploring the space of different structures, but there are there are solutions to that which are using statistics, but not in machine learning. Karl Friston: And i'm thinking of things like basic model reduction, so if you know the functional form. Karl Friston: of your ears and your priors and you make the conceptual. Karl Friston: link between a prior and a model, then you can do your model selection and structured learning basically prior learning by. Karl Friston: frame it in terms of your different prize, you could bring to the table, and there are procedures of actually allow you to score thousands of different structures or models or priors. Karl Friston: On the basis of inversion and under under a full model, so I think there are really interesting ways forward. Karl Friston: From from the point of view that the low dimensional kinds of gentle models that we deal with which I suspect from your perspective would be regarded as some toy models on demonstration models The other issue I think is important to distract knowledge which is this. Karl Friston: over competence issue you're absolutely right that sort of variation influence because you're assuming a particular functional form for your post areas and does is typically characterized by know the competence problem, and that certainly has to be offset. Karl Friston: are accounted for when doing sort of variation in print some variation learning. Karl Friston: So, you know that that is an issue usually get resolved by structure learning by by

having some slightly less coarse grained approximate posters. Yoshua Bengio: Thank you for a. shreya ahuja: Job the person will be taking two more questions, after which will be finishing So the first question is is this thermodynamic model of the brain related to JEREMY england's. shreya ahuja: dissipation driven adaptation and, if so, how is our policy selection process or microcosm of our evolutionary history. Karl Friston: Right and. Karl Friston: challenge and questions that inform questions and so. Karl Friston: On one level. Karl Friston: There is no connection between a variation free energy and the thermodynamic advantage of the consummate jamming and. Karl Friston: is dealing with your the phrase free energy just inherits from the functional form of the the objective functions and. Karl Friston: However, there are in certain physical systems and you're particularly ensemble thermodynamic ensemble systems, and you can certainly see that same kind of self organization for energy minimizing behavior. Karl Friston: When you when you read the. Karl Friston: When you read those those functions as a thermodynamic free energy and and the kind of problems that JEREMY England is dealing with is sort of. Karl Friston: A similar problem, but using a very different kind of mechanics in terms of thermodynamic free energies where he's trying to understand the emergence of self organization. Karl Friston: And how you you pay the the energetic costs of energy production that is normally associated with distributed systems that maintain themselves in some kind of non equilibrium steady state. Karl Friston: And the free energy that i'm talking about them is a little bit distinctive very distinct from that is it's purely an information theory and does rest upon having a separation between. Karl Friston: The states that are encoding the beliefs and the things that one has beliefs about so you have to have. Karl Friston: A distinction between internal external states, and it is in that distinction that the free energy i'm talking about emerges However, having said that you know. Karl Friston: they're both trying to understand what are the properties of self organized states and have some steady state aspect have some engineering aspect. Karl Friston: either from the point of view of a physicist thermodynamics and non equilibrium steady states or from the point of view of a dynamic system information theorist, how do you keep. Karl Friston: Inside you're attracting set in some sets those that serve some some state space, and the second part of the question was the. Karl Friston: remind me. Yes. shreya ahuja: It was How does that bother see selection process microcosm of ideation or history. Karl Friston: Well, I think again a great question and we've already mentioned that to a certain extent, so the process. Karl Friston: Your policy selection is just an actor basic model selection what is basic model selection well it's just taking a number of structures are a number of

models of hypotheses scoring them. Karl Friston: In terms of that module hybrid or evidence or an evidence bouncers patient free energy and then just selecting the one with the most evidence and so that's how policy selection works in. Karl Friston: In the simulations that I showed you and and that's how basically model selection works in model comparison and statistics and refining the structure various models. Karl Friston: And, and there are many people who think, particularly those subscribers of universal Darwinism and and people who have a Beige and filtering take on things like replicated dynamics. Karl Friston: In vertical biology, there are people who think that natural selection is just this process it's just the process of. Karl Friston: Implementing basic model selection over evolutionary timescale and it's an interesting game to work out well what's the hypothesis and what's the evidence. Karl Friston: And if you sort of think through it usually transpires that that basically the the model becomes a phenotype and the hypothesis is. Karl Friston: evolutions hypothesis is this phenotype a good fit for for me, the environment and then i'm going to select those that have the greatest module likelihood the greatest model evidence, what does that look like I don't simply mean that those things that are more likely to. Karl Friston: exist in the sense of they are characterized by this set of exchanges with the environment that observations of their outcomes. Karl Friston: are going to be are going to be retained so it's it's a bit has has that tautology and the natural selection hands but nicely now articulated in terms of. Karl Friston: natural selection operating to maximize an objective function says put a TD ology on natural selection, whether teleology could be written down. Karl Friston: As a free energy function and people have done that people like I think Stephen frank have shown that under some simplifying assumptions replicated dynamics is just a base income filter. Karl Friston: There are other people who take a slightly more heuristic approach from the point of view of universal Darwinism. Karl Friston: and Karl Friston: Again sort of celebrate the sort of the implicit tautology by associating adaptive fitness with just this evidence that I exist, which is the, which is the model evidence. shreya ahuja: Thank you so much, Dr Kristen i'm going to read the last question for you. shreya ahuja: It says does language also not create novel ball possibilities through a mechanism, like the adjustment possible, as suggested by Stuart Kaufman. shreya ahuja: Perhaps prompting a positive feedback loop that makes the notion of accuracy of control impossible not to mention necessitating probability measures and expanding spaces spans something a species problem. Karl Friston: i'm not going to provide a competency. Karl Friston: answer to that. Karl Friston: To that question. Karl Friston: And, and I also notice that it will follow on question and i've seen that evolution is the only and

selection is absolutely not yeah. Karl Friston: It had the converse so young, that you get the selected processor everywhere and all sorts of time scales and and also mindful of one of my mentors cherry edelman bringing. Karl Friston: Your own group selection to the table as an instance of this selective selective process as well, so you'll see this in your development, you see, you see this, so you can see, the selection of. Karl Friston: Non redundant sign up says June sleep in terms of stuff to kind of stay so positive Julia to know knee which can be cast as a complexity minimizing. Karl Friston: device that does a kind of structured learning every day so it's not just natural selection in evolutionary talking about any selected process based upon marginal liking, it is you know. Karl Friston: can be put on in that camp and the question about the the uncountable infinity inflating the sort of finite infinity of language was that the question and does that not make everything unpredictable but was other spiritual, the question is it the chairperson. Karl Friston: see what i've got that correct or. shreya ahuja: Should unmute the speaker one second give me one. Kripa Rajshekhar: hello, can you hear me. Karl Friston: Yes, I can oh. Kripa Rajshekhar: Thank you very much your answers have been very helpful and I think it's what you were saying. Kripa Rajshekhar: But maybe less sophisticated in terms of you know my understanding of the adjacent possible is that you invent a word and that word of let's say a blockchain or bitcoin then creates possibilities for a certain set of actions that could then create yet new possibilities. Karl Friston: That would. Kripa Rajshekhar: Previously, be impossible. Kripa Rajshekhar: And that, because of this exciting space, you have the notion of positive feedback and it connects to the whole energy issue, which is. Kripa Rajshekhar: can be really in the context of positive feedback is the notion of control, when you have that type of singularity does this all sort of just become more or less impossible, and we need new tools in that situation and language, being an example. Karl Friston: Now is a great example of one of the great challenges in terms of getting your proper. Karl Friston: Natural language understanding under the Hood in in sort of machine learning and a great question and I think there's a practical answer to that question is that, certainly in the sense of I came across this notion of dial operators in computation theory and the replication. Karl Friston: That ensues from that and. Karl Friston: The way that that doesn't. Karl Friston: offer. Karl Friston: A problem from the point of view of building a good judge of models to understand say language is again it's as simple and imperative to keep things simple so every time you induce another. Karl Friston: Another level or layer or composition. Karl Friston: or sequence of a sequence of a sequence, and also hash code or zip codes zip codes every time you do that you're juicing more degrees of freedom, so you can only do that. Karl

Friston: If the by introducing there's extra degrees of freedom you more than pay the price in terms of increasing the accuracy and, of course, at some point the predictability or the thing you're trying to. Karl Friston: supply an accurate explanation for will make that impossible so you're going to go up to as complex as possible but at some point going any further you're going to start to have an oval express security model. Karl Friston: That is going to not generalize it's going to overfit and, therefore, if you've got the right compare complexity penalty and play it will start a simplified so. Karl Friston: itself again so from the perspective, the infinite possibilities uncommon attacks that one, there is a 40 by things like language. Karl Friston: is not a problem from the point of view of trying to understand how this person or how this culture is generating language and and, indeed, one can then sort of tell very simple arguments, where you. Karl Friston: can understand this variation of treatment of self organization in terms of just making the world more and more predictable. Karl Friston: How do you do that, where you learn to speak the same language as your specifics. Karl Friston: And indeed you've learned to speak the same language as your environment you start to build your environment, so it supplies and ambiguous simple things that you predict like edges, so do you think by. Karl Friston: And you know I entertainers in the world, but language I think is a beautiful beautiful example of this, it it just enables us to predict each other and. Karl Friston: The question about tools, then well So what about this yeah beautiful mathematical. Karl Friston: recursive aspect that in principle means if you go out there into hyperspace with with various operators or functions and representations graphical models or stick baking processes, and I think that that kind of math and that kind of, if you like. Karl Friston: expressive it may be, I don't know how, but I would he may be the solution the tool that solves what we were talking about. Karl Friston: With the previous challenging questions about sort of the structure learning exploring the space of models, it may be that they are having that kind of calculus at hand. Karl Friston: allows you in a reasonably structured way to explore extensions to your generative model, and if you can do that, then we already have the. Karl Friston: The mechanics to score this new one, and if it passes the Marks and the evidence increases of the free energy free energy increases, then we retain it. Karl Friston: If not, we revert back again so it may be i'm not quite sure what kind of calculus you know compilation failure, you had in mind, but if he is all the saw that that. Karl Friston: brings new, more complicated architectures to the table or mappings to the table of functions to the. Karl Friston: To the table, then in a principled way, then it might hold the key to how we explore model spaces and and do structured learning and I think that that's again it comes back to the

earlier question that we were wrestling with in terms of you know. Karl Friston: you're avoiding overfitting and you're getting the right conditioning on context, like. Karl Friston: it's not the kind of calculus is that we had in mind. Kripa Rajshekhar: But while we're working in the area of innovation, and you know there's like a variant of the pole. Kripa Rajshekhar: pole you earn sort of sampling method which is related to the digital a process, but you want to actually have some notion of a possibility that then colors the types of future possibilities that might become possible again i'm using possible too much, but you know what i'm saying. Karl Friston: I think so. Karl Friston: I probably don't actually have probably. Kripa Rajshekhar: Thank you very much for your answer, though. Karl Friston: But I mean certainly your hiring process is, I think a really nice example in terms of how far you want to go and extending them across that that's the problem that. Karl Friston: People non parametric bays are trying you're trying to resolve and it may be really important to solve that to get so natural language processing. Karl Friston: Your as fluid, as it can be. Thank you. Okay. Anna Choromanska: Thanks a lot caught it for very exciting to. Anna Choromanska: Talk Thank you so much, everybody for questions, I hope, we managed to cover most of the questions if not. Anna Choromanska: Well, at these discussions that. Anna Choromanska: We had on the question and answers, as well as chat. Anna Choromanska: Again, thank you so much for staying with us so late, because we are on different continents and we do hope to see you in person in New York. Anna Choromanska: Sometime soon. Karl Friston: Thank you very much it's an honor to speak to you, it was a pleasure and so great questions i'm gonna have to go where I go to have a pseudo think about them and see if I could have thought of it better answers. Anna Choromanska: Thanks so much Thank you so much, once again, thanks everybody Thank you so much. Karl Friston: Hello. bye.